Quantum Fluctuations in the Neutrino Sector: A Perturbative Framework

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Introduction

Neutrino physics stands at the frontier of modern particle theory, where subtle quantum phenomena reveal profound truths about the structure of the universe. Among these, the oscillation of neutrinos—the transformation between electron, muon, and tau flavors—has reshaped our understanding of mass generation and CP violation. Despite their elusive nature, neutrinos provide a unique window into physics beyond the Standard Model.

This project presents a numerical and analytical study of **neutrino mass matrix perturbations**, using global-fit data from *NuFIT* as the experimental foundation. The analysis explores how infinitesimal deviations in the neutrino mass matrix influence key observables: the PMNS mixing angles, the CP-violating phase δ_{CP}, and the Jarlskog invariant J.

By implementing a Monte Carlo perturbation framework, the code systematically introduces small symmetric fluctuations in the Majorana mass matrix, diagonalizes the resulting structure, and tracks the evolution of physical parameters across varying perturbation scales. The resulting distributions and correlations shed light on the stability and sensitivity of the neutrino sector under theoretical perturbations.

Beyond reproducing established results, this framework opens a pathway for exploring deeper symmetries and texture patterns that may underlie the observed neutrino mixing, bridging the gap between phenomenology and theoretical model-building.

Theoretical Background

1. Neutrino Oscillations

Neutrino oscillations represent one of the most striking manifestations of quantum mechanics in particle physics. They describe the phenomenon whereby a neutrino created in a definite flavor eigenstate—such as an electron, muon, or tau neutrino—can later be detected as a different flavor. This flavor transition is inherently quantum in nature and arises because the flavor eigenstates that participate in weak interactions are not identical to the mass eigenstates that propagate through space.

Mathematically, the flavor and mass eigenstates are related through the **Pontecorvo–Maki–Nakag (PMNS) matrix**, a complex unitary 3×3 matrix that encodes the mixing among the three neutrino generations. Each element of this matrix determines the amplitude for a neutrino of a given mass state to appear as a specific flavor. As neutrinos travel, the mass eigenstates—each characterized by slightly different masses—evolve with different phases. The interference of these evolving components produces the observed oscillatory pattern in flavor probabilities.

The PMNS matrix contains three mixing angles $(\theta_{12}, \theta_{13}, \theta_{23})$ and two mass-squared differences $(\Delta m^2_{21}, \Delta m^2_{3l})$, which govern the oscillation frequencies. The angle θ_{12} dominates solar neutrino oscillations, θ_{23} controls atmospheric neutrino transitions, and θ_{13} is primarily associated with reactor neutrinos. The values of $\Delta m^2_{21} = m^2_2 - m^2_1$ and $\Delta m^2_{3l} = m^2_3 - m^2_l$ (with $l = 1$ or 2) determine the oscillation wavelengths and depend on whether the neutrino mass ordering is normal or inverted.

The existence of neutrino oscillations provides irrefutable evidence that neutrinos have non-zero masses, marking a profound extension of the Standard Model. Precise knowledge of the PMNS matrix is crucial not only for describing oscillation behavior but also for investigating CP violation in the lepton sector. The matrix's complex phase introduces asymmetries between neutrino and antineutrino oscillations, connecting microscopic quantum effects to the macroscopic matter–antimatter imbalance in the universe.

Hence, determining the PMNS parameters with high precision is essential for studying how perturbations in the mass matrix affect the Dirac phase δ_{CP} and the Jarlskog invariant, both key quantities in quantifying CP violation.

2. CP Violation in the Lepton Sector

CP violation refers to the breakdown of the combined symmetry of charge conjugation (C)—which transforms particles into their antiparticles—and parity (P)—which reflects spatial coordinates. In a CP-symmetric universe, the laws of physics would treat matter and antimatter identically; however, experiments have shown small but significant deviations from this symmetry. CP violation is therefore indispensable for explaining why the observable universe contains vastly more matter than antimatter.

In the lepton sector, CP violation manifests through the **Dirac phase** δ_{CP} in the PMNS matrix. This complex phase induces measurable differences in the oscillation probabilities between neutrinos and antineutrinos. For instance, the transition probability $P(\nu_\mu \to \nu_e)$ may differ from its CP-conjugate process $P(\bar{\nu}_\mu \to \bar{\nu}_e)$, a difference known as CP asymmetry. These effects depend on both the neutrino's energy and the propagation distance, and they are actively sought in modern long-baseline experiments such as T2K, NOvA, and DUNE.

To quantify CP violation independently of parameterization, one defines the **Jarlskog invariant** J, a rephasing-invariant quantity constructed from products of PMNS matrix elements:

$$J = \mathrm{Im}(U_{e1}U_{\mu 2}U_{e2}^{*}U_{\mu 1}^{*}).$$

This invariant vanishes if and only if CP symmetry is conserved. Its non-zero value directly measures the magnitude of CP violation, and any perturbation in the PMNS matrix—whether in its elements, mixing angles, or complex phase—alters J. Consequently, the Jarlskog invariant serves as a sensitive diagnostic tool for studying how theoretical perturbations or experimental uncertainties affect CP-violating observables. Understanding these sensitivities is central to assessing the stability of theoretical models of neutrino mixing and their implications for the early universe.

3. Leptogenesis

The phenomenon of **leptogenesis** offers an elegant cosmological mechanism that connects neutrino physics to one of the deepest mysteries in cosmology: the dominance of matter over antimatter. In the early universe, heavy right-handed (Majorana) neutrinos—hypothetical particles predicted by many extensions of the Standard Model—could have decayed out of thermal equilibrium, producing a slight excess of leptons over antileptons. This **lepton asymmetry** was later converted into a **baryon asymmetry** through non-perturbative electroweak processes known as sphaleron transitions, which violate lepton and baryon number conservation but preserve $B - L$.

This mechanism is deeply intertwined with the **seesaw model**, which explains the tiny observed neutrino masses as a consequence of the heavy mass scale of right-handed neutrinos. The decay of Majorana neutrinos inherently violates lepton number, and their CP-violating decay amplitudes naturally generate the required asymmetry. The relationship $Y_B \sim -Y_L$ expresses how a preexisting lepton asymmetry Y_L gives rise to the observed baryon asymmetry Y_B.

The importance of flavor effects cannot be overstated: depending on whether the asymmetry originates in the electron, muon, or tau sector, the final baryon asymmetry may differ substantially. Consequently, precise modeling of neutrino mixing parameters and CP-violating phases is indispensable for predicting the outcome of leptogenesis. Perturbation analyses of the PMNS matrix and the underlying mass matrices allow physicists to explore how small deviations in parameters might influence the resulting lepton asymmetry, thus offering insight into the flavor sensitivity of cosmological matter–antimatter generation.

4. Mass Matrices and Perturbation Analysis

At the theoretical core of neutrino physics lies the **mass matrix**, which encodes the fundamental constants governing neutrino masses and mixings. Depending on whether neutrinos are Dirac or Majorana particles, the structure of this matrix and its symmetries differ significantly. For Dirac neutrinos, lepton number is conserved, while for Majorana neutrinos, lepton number is violated—a key requirement for leptogenesis.

Studying perturbations of the mass matrix allows researchers to understand the stability of CP-violating observables such as δ_{CP} and the Jarlskog invariant. Even small modifications to the elements of the mass matrix—arising from higher-order corrections, symmetry breakings, or experimental uncertainties—can propagate into significant shifts in these observables. Quantifying such sensitivities is crucial for testing the robustness of theoretical models and ensuring that experimentally determined values of CP violation are not artifacts of parameter fine-tuning.

Modern approaches employ **Monte Carlo simulations** to systematically explore how random perturbations in the mass matrix affect measurable quantities. Through this method, one can map the statistical behavior of the resulting δ_{CP} and J values, providing a quantitative assessment of model resilience under small deviations. In this sense, perturbation analysis serves as a bridge between abstract theoretical formulations and their phenomenological implications, directly informing both model building and experimental strategy.

Project Significance

Understanding the stability of neutrino mixing parameters under small perturbations is of profound theoretical and phenomenological importance. The observed pattern of neutrino oscillations implies the existence of a highly non-trivial mass structure that remains only partially understood within the Standard Model framework. Despite impressive experimental progress, uncertainties in the CP-violating phase δ_{CP} and the absolute neutrino mass scale still limit our ability to identify the true underlying symmetries of the lepton sector.

This project addresses one of the key open questions: *How robust are the observed neutrino mixing parameters against small perturbations in the mass matrix?* By analyzing this sensitivity, the research provides insights into whether the current parametrization of the PMNS matrix can emerge naturally from small deviations in underlying symmetries, or whether it requires fine-tuning.

The significance of this analysis lies in three main aspects:

- It bridges theoretical neutrino model-building with numerical perturbation studies, offering a unified computational framework.
- It quantifies the resilience of CP-violating observables such as δ_{CP} and J against microscopic fluctuations, providing constraints on viable theoretical textures.
- It supports ongoing experimental efforts (T2K, NOvA, DUNE) by evaluating how measurement uncertainties could reflect intrinsic theoretical instabilities.

Through this combination of theory and simulation, the project contributes to a deeper understanding of how small-scale quantum fluctuations may shape large-scale features such as leptonic CP violation and, ultimately, the matter–antimatter asymmetry of the universe.

Objectives

The primary objective of this project is to develop a robust computational and theoretical framework for analyzing how perturbations in the neutrino mass matrix influence measurable physical quantities, particularly the CP-violating phase δ_{CP} and the Jarlskog invariant J. By implementing a systematic perturbative approach, the study aims to establish quantitative links between microscopic model variations and macroscopic observables.

The objectives of the project can be summarized as follows:

1. **Construct** a reliable numerical scheme capable of generating and diagonalizing perturbed Majorana mass matrices using Monte Carlo sampling.
2. **Evaluate** how small random perturbations affect neutrino mixing parameters and CP-violating quantities derived from the PMNS matrix.
3. **Quantify** the stability and sensitivity of δ_{CP} and J under controlled theoretical fluctuations.
4. **Identify** potential correlations between perturbation strength and deviations from experimental best-fit parameters, assessing model robustness.
5. **Provide** a reproducible computational framework that can be extended to test new neutrino mass textures and theoretical symmetries.

Together, these objectives form the foundation for an in-depth exploration of the relationship between theoretical perturbations and observable neutrino phenomena, helping to clarify whether the structure of the PMNS matrix is a stable feature of lepton physics or a product of fine-tuned dynamics.

Core Concept

The core concept of this project lies in treating the neutrino mass matrix as a dynamic, perturbable entity whose microscopic fluctuations can propagate into observable quantities. Within this framework, the neutrino mixing parameters—encoded in the PMNS matrix—are not fixed constants but emergent properties arising from the underlying mass structure. By introducing controlled perturbations to the theoretical mass matrix, the analysis aims to uncover how sensitive these emergent properties are to small deviations in the fundamental parameters.

The project adopts a **perturbative Monte Carlo framework** that systematically generates large ensembles of slightly modified mass matrices around a given reference configuration consistent with current global-fit data. Each perturbed matrix is subsequently diagonalized to extract the corresponding eigenvalues (masses) and eigenvectors (mixing parameters). Statistical evaluation of these ensembles provides insight into the spread, correlations, and robustness of key observables such as θ_{12}, θ_{13}, θ_{23}, δ_{CP}, and J.

This approach combines analytical structure with numerical exploration, offering a bridge between abstract theoretical formulations and computational phenomenology. By analyzing the emergent statistical distributions of the mixing parameters, the framework reveals whether the observed neutrino behavior could arise naturally from small quantum fluctuations, or whether additional symmetries and constraints are required to stabilize the PMNS structure.

Methodology and Tools

The methodological foundation of this project integrates analytical reasoning with numerical experimentation to investigate the impact of mass matrix perturbations on neutrino observables. The analysis is structured around the following key computational components:

- **Monte Carlo Sampling:** Random perturbations are applied to the base Majorana neutrino mass matrix to explore a wide statistical space of possible microscopic deviations. Each perturbation is drawn from a controlled probability distribution, ensuring that fluctuations remain within physically realistic limits.
- **Matrix Diagonalization:** Every generated mass matrix is diagonalized to ob-

tain the corresponding eigenvalues and eigenvectors. The eigenvalues represent the physical neutrino masses, while the eigenvectors define the flavor-mixing parameters that form the PMNS matrix.

- **Parameter Extraction:** From the diagonalized matrices, the mixing angles $(\theta_{12}, \theta_{13}, \theta_{23})$ and CP-violating phase δ_{CP} are extracted using standard parameterization conventions. The Jarlskog invariant J is then computed as a derived quantity to quantify CP violation.

- **Statistical Analysis:** The procedure is repeated over thousands of Monte Carlo iterations to obtain statistically significant distributions of all observables. These distributions are analyzed to determine mean values, variances, and correlations among key parameters.

The computational implementation is carried out using **Python**, with core numerical operations handled by `NumPy` and `SciPy` libraries. Visualization and statistical post-processing are performed using `Matplotlib` and `Pandas`. The resulting framework provides both precision and scalability, enabling high-volume simulations without compromising analytical rigor.

Mathematical Framework and Perturbation Analysis

The neutrino mass matrix in the flavor basis can be expressed as:

$$M_\nu = U^* \,\mathrm{diag}(m_1, m_2, m_3)\, U^\dagger, \tag{1}$$

where U is the PMNS matrix and m_i are the neutrino mass eigenvalues. In the Monte Carlo simulation, random perturbations ΔM_ν are introduced to this matrix:

$$M'_\nu = M_\nu + \Delta M_\nu, \tag{2}$$

with ΔM_ν generated from a complex random ensemble satisfying

$$\langle \Delta M_{\nu,ij} \rangle = 0, \qquad \langle |\Delta M_{\nu,ij}|^2 \rangle = \sigma_M^2, \tag{3}$$

where σ_M controls the perturbation amplitude.

The perturbed matrix M'_ν is then diagonalized numerically:

$$U'^T M'_\nu U' = \mathrm{diag}(m'_1, m'_2, m'_3). \tag{4}$$

From U' we extract the three mixing angles and the CP phase using the standard parametrization:

$$s_{13} = |U'_{e3}|, \tag{5}$$

$$t_{12} = \frac{|U'_{e2}|}{|U'_{e1}|}, \tag{6}$$

$$t_{23} = \frac{|U'_{\mu 3}|}{|U'_{\tau 3}|}, \tag{7}$$

$$\delta_{CP} = \arg\left(\frac{U'_{e1} U'_{\mu 3} U'^*_{e3} U'^*_{\mu 1}}{|U'_{e1} U'_{\mu 3} U'^*_{e3} U'^*_{\mu 1}|} \right). \tag{8}$$

To ensure physical consistency, the eigenvalues are constrained to be positive and the resulting U' must satisfy unitarity:

$$U'^\dagger U' = I + \mathcal{O}(\epsilon_U), \tag{9}$$

where the deviation parameter is defined as

$$\epsilon_U = \|U'^\dagger U' - I\|_F. \tag{10}$$

In practice, $\epsilon_U \lesssim 10^{-10}$ for numerical stability.

For each realization, the squared mass differences are computed as:

$$\Delta m^2_{21} = m'^2_2 - m'^2_1, \qquad \Delta m^2_{31} = m'^2_3 - m'^2_1, \tag{11}$$

and compared to experimental constraints:

$$\Delta m^2_{21} \approx 7.5 \times 10^{-5}\,\mathrm{eV}^2, \tag{12}$$

$$|\Delta m^2_{31}| \approx 2.5 \times 10^{-3}\,\mathrm{eV}^2. \tag{13}$$

The probability of flavor transition between neutrino types is given by:

$$P_{\alpha\to\beta} = \delta_{\alpha\beta} - 4\sum_{i>j} \mathrm{Re}(U_{\alpha i}U^*_{\beta i}U^*_{\alpha j}U_{\beta j}) \sin^2\left(\frac{\Delta m^2_{ij}L}{4E}\right) + 2\sum_{i>j} \mathrm{Im}(U_{\alpha i}U^*_{\beta i}U^*_{\alpha j}U_{\beta j}) \sin\left(\frac{\Delta m^2_{ij}L}{2E}\right), \tag{14}$$

where L is the baseline distance and E is the neutrino energy.

During the Monte Carlo cycles, each perturbed matrix yields a different set of parameters $\{\theta^{(k)}_{12}, \theta^{(k)}_{13}, \theta^{(k)}_{23}, \delta^{(k)}_{CP}\}$. The stability of the mixing pattern is analyzed statistically:

$$\text{Stability Index} = \sqrt{\sum_{ij}\left(\frac{\sigma_{\theta_{ij}}}{\langle\theta_{ij}\rangle}\right)^2 + \left(\frac{\sigma_\delta}{\langle\delta_{CP}\rangle}\right)^2}. \tag{15}$$

If this index remains small for increasing σ_M, the model is said to be *perturbatively stable*, implying that the observed PMNS structure emerges naturally from the underlying mass texture.

```
# Example: Monte Carlo perturbation loop
for k in range(N):
    M_pert = M_nu + np.random.normal(0, sigma_M, M_nu.shape)
    evals, evecs = np.linalg.eigh(M_pert)
    U = normalize(evecs)
    angles[k], delta[k] = extract_pmns_parameters(U)
```

These Monte Carlo and analytical analyses together provide a comprehensive view of how microscopic perturbations in M_ν propagate into macroscopic observables such as oscillation probabilities, CP violation, and neutrino mass ordering.

Data Collection and Result Visualization

After performing N Monte Carlo realizations of random perturbations in the neutrino mass matrix M_ν, the resulting set of diagonalized PMNS parameters $\{\theta_{12}^{(k)}, \theta_{13}^{(k)}, \theta_{23}^{(k)}, \delta_{CP}^{(k)}\}$ are statistically analyzed.

The mean and standard deviation of each mixing angle are defined as:

$$\langle\theta_{ij}\rangle = \frac{1}{N}\sum_{k=1}^{N}\theta_{ij}^{(k)}, \qquad \sigma_{\theta_{ij}} = \sqrt{\frac{1}{N-1}\sum_{k=1}^{N}(\theta_{ij}^{(k)} - \langle\theta_{ij}\rangle)^2}. \tag{16}$$

Similarly, the distribution of the Dirac phase δ_{CP} and Jarlskog invariant J can be used to quantify CP-violation stability under perturbations:

$$J^{(k)} = \tfrac{1}{8}\sin\left(2\theta_{12}^{(k)}\right)\sin\left(2\theta_{13}^{(k)}\right)\sin\left(2\theta_{23}^{(k)}\right)\cos\left(\theta_{13}^{(k)}\right)\sin\left(\delta_{CP}^{(k)}\right), \tag{17}$$

$$\langle J\rangle = \frac{1}{N}\sum_{k=1}^{N}J^{(k)}, \qquad \sigma_J = \sqrt{\frac{1}{N-1}\sum_{k=1}^{N}(J^{(k)} - \langle J\rangle)^2}. \tag{18}$$

The numerical distributions of δ_{CP} are compared to theoretical expectations by fitting a Gaussian:

$$P(\delta_{CP}) = \frac{1}{\sqrt{2\pi\sigma_\delta^2}}\exp\left[-\frac{(\delta_{CP} - \langle\delta_{CP}\rangle)^2}{2\sigma_\delta^2}\right]. \tag{19}$$

To evaluate the stability of unitarity in each Monte Carlo realization, the deviation from identity of the reconstructed PMNS matrix $U^{(k)}$ is computed as:

$$\epsilon_U^{(k)} = \left\|U^{(k)\dagger}U^{(k)} - I\right\|_F, \tag{20}$$

where $\|\cdot\|_F$ denotes the Frobenius norm. The ensemble average $\langle\epsilon_U\rangle$ serves as a quantitative diagnostic of the model's internal consistency.

Statistical correlations between the mixing parameters can also be assessed through the Pearson coefficient:

$$\rho_{ij,kl} = \frac{\langle (\theta_{ij} - \langle \theta_{ij} \rangle)(\theta_{kl} - \langle \theta_{kl} \rangle) \rangle}{\sigma_{\theta_{ij}} \sigma_{\theta_{kl}}}, \tag{21}$$

which identifies whether perturbations affecting one angle propagate coherently to others.

The results are finally visualized by generating histograms and correlation plots using Python:

```
plt.hist(theta_12_values, bins=50)
plt.xlabel(r"$\theta_{12}$ (degrees)")
plt.ylabel("Frequency")
plt.title("Distribution of $\theta_{12}$ under matrix perturbations")
plt.show()
```

These analyses reveal whether the experimentally observed PMNS parameters are statistically stable under microscopic deformations of M_ν. A narrow distribution of δ_{CP} or J indicates robustness of the CP-violating phase, whereas broad or multimodal structures may point to hidden symmetries or fine-tuning in the neutrino sector.

Correlation and Covariance Structure of the Perturbed Mass Matrix

Beyond simple perturbative stability, a crucial aspect of the analysis involves studying how the elements of M_ν co-vary under random perturbations. This is captured by the empirical covariance matrix:

$$\Sigma_{ij,kl} = \langle (M'_{\nu,ij} - \langle M'_{\nu,ij} \rangle)(M'_{\nu,kl} - \langle M'_{\nu,kl} \rangle)^* \rangle, \tag{22}$$

where the brackets $\langle \cdot \rangle$ denote the Monte Carlo ensemble average. This matrix quantifies the degree of correlation between perturbations applied to different entries of M_ν.

The normalized correlation coefficient between two elements is defined as:

$$\rho_{ij,kl} = \frac{\mathrm{Re}(\Sigma_{ij,kl})}{\sqrt{\Sigma_{ij,ij}\,\Sigma_{kl,kl}}}, \qquad -1 \leq \rho_{ij,kl} \leq 1. \tag{23}$$

If the perturbations are uncorrelated, $\rho_{ij,kl} \simeq 0$ for $(ij) \neq (kl)$. However, during diagonalization, numerical correlations can arise due to the orthogonality constraints on the eigenvectors. Such correlations can propagate into the extracted PMNS parameters, leading to collective shifts in θ_{ij} and δ_{CP}.

To visualize these dependencies, the covariance matrix of the extracted angles is computed:

$$C_{\alpha\beta} = \langle(\theta_\alpha - \langle\theta_\alpha\rangle)(\theta_\beta - \langle\theta_\beta\rangle)\rangle, \qquad \alpha, \beta \in \{12, 13, 23, \delta\}. \tag{24}$$

From this matrix we extract both the standard deviations and the correlation coefficients:

$$\rho_{\alpha\beta} = \frac{C_{\alpha\beta}}{\sqrt{C_{\alpha\alpha}C_{\beta\beta}}}. \tag{25}$$

The eigen-decomposition of C reveals the principal directions of uncertainty propagation:

$$C = R\,\mathrm{diag}(\lambda_1, \lambda_2, \lambda_3, \lambda_4)\,R^T, \tag{26}$$

where the eigenvalues λ_i represent the variances along orthogonal axes in parameter space. A dominance of one large eigenvalue indicates that the system's instability is driven by a single correlated fluctuation mode.

```
# Covariance and correlation of mixing angles
C = np.cov(np.array([theta12, theta13, theta23, deltaCP]))
corr = np.corrcoef(np.array([theta12, theta13, theta23, deltaCP]))
eigvals, eigvecs = np.linalg.eigh(C)
```

Finally, a compact measure of global correlation strength can be defined as:

$$\mathcal{R} = \frac{1}{n(n-1)} \sum_{\alpha\neq\beta} |\rho_{\alpha\beta}|, \tag{27}$$

where $n = 4$ is the number of parameters. A small value of $\mathcal{R}$ indicates statistical independence between the mixing parameters, suggesting that the PMNS structure is dynamically decoupled. Conversely, a large $\mathcal{R}$ implies strong mutual dependencies and possible fine-tuning within the neutrino sector.

These analyses, both analytical and numerical, together build a comprehensive picture of how random fluctuations in the underlying mass matrix translate into structured correlations among the observable parameters.

Expected Results and Extracted Observables

Based on the Monte Carlo perturbation framework applied to the neutrino mass matrix, we anticipate the following features:

1. Stability of Mixing Angles

The distributions of the extracted PMNS angles $\theta_{12}, \theta_{13}, \theta_{23}$ are expected to cluster around their experimentally measured central values:

$$\langle \theta_{ij} \rangle = \frac{1}{N} \sum_{k=1}^{N} \theta_{ij}^{(k)}, \quad \sigma_{\theta_{ij}} = \sqrt{\frac{1}{N-1} \sum_{k=1}^{N} (\theta_{ij}^{(k)} - \langle \theta_{ij} \rangle)^2}. \tag{28}$$

Here, N denotes the number of Monte Carlo samples. These quantities quantify the robustness of the observed mixing parameters under microscopic perturbations of the Majorana mass matrix M_ν.

2. Mass-Squared Differences and Hierarchies

We expect the mass-squared differences to remain consistent with the normal or inverted ordering:

$$\Delta m_{21}^2 = m_2^2 - m_1^2, \qquad \Delta m_{31}^2 = m_3^2 - m_1^2. \tag{29}$$

Their statistical spreads $\sigma_{\Delta m^2}$ indicate the sensitivity of oscillation frequencies to perturbations.

3. CP Violation Sensitivity

The Jarlskog invariant J and Dirac CP phase δ_{CP} are expected to exhibit distributions:

$$J = c_{12}c_{23}c_{13}^2 s_{12}s_{23}s_{13}\sin\delta_{CP}. \tag{30}$$

Non-zero variance in J reflects how microscopic deviations in M_ν can influence observable CP asymmetry.

4. Expected Visualizations

Histograms and correlation plots (constructed from the Monte Carlo outputs) should reveal Gaussian-like distributions for the mixing angles and CP phase. Correlation matrices can expose non-trivial dependencies:

$$\rho(\theta_i, \theta_j) = \frac{\mathrm{Cov}(\theta_i, \theta_j)}{\sigma_{\theta_i}\sigma_{\theta_j}}. \tag{31}$$

Conclusions

From the computational framework and Monte Carlo simulations, several conclusions emerge:

- The PMNS mixing angles remain statistically stable under small symmetric perturbations of M_ν, confirming the robustness of the experimentally observed values.
- The mass-squared differences Δm^2_{21} and Δm^2_{31} are consistent with the chosen mass ordering, with minor variations due to the perturbation scale.
- CP-violating observables, especially J and δ_{CP}, are sensitive indicators of small deviations in the neutrino mass matrix, making them excellent probes for theoretical model testing.
- The Monte Carlo framework validates the methodology of connecting theoretical perturbations with observable quantities in neutrino physics.

Future Developments

Potential extensions and improvements of this study include:

1. Incorporating **non-Gaussian perturbations** to explore more exotic flavor structures.
2. Extending the framework to **Dirac neutrino scenarios** and comparing the impact on CP violation.
3. Including **renormalization group effects** to study the evolution of M_ν from high energy scales down to low energies.
4. Implementing advanced **statistical techniques** (e.g., Bayesian inference, Markov Chain Monte Carlo) to refine parameter estimation.
5. Combining numerical results with **analytical constraints** from flavor symmetries to identify viable texture patterns in M_ν.

Appendix: Project Code and Usage

1. Full Project Code

```
import numpy as np


# Define base neutrino mass matrix M_nu (Majorana, symmetric)
M_base = np.array([[a, b, c],
                   [b, d, e],
                   [c, e, f]], dtype=complex)


# Monte Carlo perturbation function
def delta_M():
    epsilon = np.random.normal(0, sigma, size=(3,3)) + 1j*np.random.normal(0, sigma,
    epsilon = (epsilon + epsilon.T)/2  # ensure symmetry
    return epsilon


# Extract mixing angles from eigenvectors
def extract_angles(U):
    # Standard PMNS parametrization
    theta_13 = np.arcsin(abs(U[0,2]))
    theta_12 = np.arctan(abs(U[0,1])/abs(U[0,0]))
    theta_23 = np.arctan(abs(U[1,2])/abs(U[2,2]))
    delta = np.angle(U[0,2])
    return theta_12, theta_13, theta_23, delta


# Monte Carlo sampling
N_samples = 10000
theta_12_values = []
```

```
theta_13_values = []

theta_23_values = []

delta_values = []


for k in range(N_samples):

    M_prime = M_base + delta_M()

    eigvals, eigvecs = np.linalg.eigh(M_prime)

    theta_12, theta_13, theta_23, delta = extract_angles(eigvecs)

    theta_12_values.append(theta_12)

    theta_13_values.append(theta_13)

    theta_23_values.append(theta_23)

    delta_values.append(delta)


# Mass-squared differences

m1, m2, m3 = eigvals

Delta_m21_sq = m2**2 - m1**2

Delta_m31_sq = m3**2 - m1**2


# Jarlskog invariant

c12, c13, c23 = np.cos(theta_12), np.cos(theta_13), np.cos(theta_23)

s12, s13, s23 = np.sin(theta_12), np.sin(theta_13), np.sin(theta_23)

J = c12*c23*c13**2*s12*s23*s13*np.sin(delta)


# Statistical aggregation

theta_12_mean = np.mean(theta_12_values)

theta_12_std  = np.std(theta_12_values, ddof=1)


# Results can be saved as CSV or plotted

# Example CSV output:
```

```
import pandas as pd
df = pd.DataFrame({
    "theta_12": theta_12_values,
    "theta_13": theta_13_values,
    "theta_23": theta_23_values,
    "delta": delta_values
})
df.to_csv("neutrino_MC_results.csv", index=False)
```

2. How to Run the Code

- Ensure Python 3.x is installed.
- Required libraries: `numpy`, `pandas`, `matplotlib`.
- Modify the base mass matrix entries a, b, c, d, e, f according to your input data.
- Set the perturbation scale σ and number of Monte Carlo samples $N\_samples$.
- Run the script; the results will be saved in CSV format and can be plotted for analysis.

3. Notes and Practical Tips

- Ensure the perturbations remain small: $|\delta M_{ij}| \ll |M_{ij}|$ to maintain physical realism.
- Symmetrize the perturbation matrix to preserve Majorana nature.
- Choose $N\_samples$ large enough (e.g., 10,000 or more) for statistical convergence.
- For visualization, histograms and correlation plots help to understand parameter distributions and sensitivities.
- CSV outputs can be used for further statistical analysis or cross-checking with experimental data.

Appendix: Full colab Code and Usage Instructions

This appendix provides the full Python code used for the neutrino mass matrix perturbation analysis, including Monte Carlo simulations, extraction of PMNS parameters, statistical analysis, and output generation. The code is designed to run in Google Colab or any Python 3 environment with the required packages: `numpy`, `scipy`, `matplotlib`, `pandas`, `tqdm`.

How to Run the Code

1. Upload the `nufit.json` file if you have it, or rely on fallback default values.
2. Set the parameters in the `Args` class if needed:
    - `n_samples`: Number of Monte Carlo samples per epsilon.
    - `epsilon_list`: List of perturbation scales.
    - `mass_matrix_type`: `'Majorana'` or `'Dirac'`.
    - `ordering`: `'normal'` or `'inverted'`.
    - `random_scale`: Scale of random perturbations.
    - `complex_phase`: Include complex phase in perturbations.
3. Run the script. Outputs will include:
    - Summary CSV: `nu_analysis_summary.csv`
    - Detailed samples CSV: `nu_analysis_detailed_samples.csv`
    - Histograms of $\Delta\delta$ and ΔJ: `nu_analysis_histograms.png`
    - Scan errorbar plot: `nu_analysis_scan.png`

Full Python Code

```
from google.colab import files

uploaded = files.upload()
```

```
class Args:
    nufit_json = 'nufit.json'
    cov_path = None
    use_cov_sampler = False
    cov_scale = 1.0
    ordering = 'normal'
    mass_matrix_type = 'Majorana'
    n_samples = 200
    epsilon_list = [1e-4, 5e-4, 1e-3, 5e-3, 1e-2]
    random_scale = 1e-3
    complex_phase = False
    out_prefix = 'nu_analysis'

args = Args()

# -*- coding: utf-8 -*-
"""
NuFIT-ready neutrino-mass perturbation analysis
- Uses NuFIT (default: v6.0) as baseline parameters if available
- Supports Majorana vs Dirac, normal/inverted ordering
- Sampling from experimental covariance (if provided) or from per-parameter uncertain
- Improved extraction of mixing angles/CP following PDG conventions
- Outputs CSV summaries, PNG figures, and detailed stats (histograms, correlations, C
"""

import numpy as np
import matplotlib.pyplot as plt
from numpy import linalg as LA
```

```
import os
import json
import logging
from scipy.stats import multivariate_normal
from tqdm import tqdm
import pandas as pd

# Logging
logging.basicConfig(level=logging.INFO, format='%(levelname)s: %(message)s')

# Utilities
def deg2rad(x_deg):
    return x_deg * np.pi / 180.0
def rad2deg(x_rad):
    return x_rad * 180.0 / np.pi
def clamp(x, lo=-1.0, hi=1.0):
    return max(lo, min(hi, x))

# Build PMNS
def build_PMNS(theta12_deg, theta13_deg, theta23_deg, delta_deg):
    th12 = deg2rad(theta12_deg)
    th13 = deg2rad(theta13_deg)
    th23 = deg2rad(theta23_deg)
    delta = deg2rad(delta_deg)
    c12 = np.cos(th12); s12 = np.sin(th12)
    c13 = np.cos(th13); s13 = np.sin(th13)
    c23 = np.cos(th23); s23 = np.sin(th23)
    e_minus_i_delta = np.exp(-1j * delta)
    e_plus_i_delta  = np.exp(1j * delta)
```

```
    U = np.array([
        [ c12*c13, s12*c13, s13 * e_minus_i_delta ],
        [ -s12*c23 - c12*s23*s13*e_plus_i_delta, c12*c23 - s12*s23*s13*e_plus_i_delta
        [ s12*s23 - c12*c23*s13*e_plus_i_delta, -c12*s23 - s12*c23*s13*e_plus_i_delta
    ], dtype=complex)
    return U

# Mass matrix builders
def build_mass_matrix_from_masses(U, masses, mass_matrix_type='Majorana'):
    mdiag = np.diag(masses)
    if mass_matrix_type.lower() == 'majorana':
        M = U.dot(mdiag).dot(U.T)
    else:
        M = U.dot(mdiag).dot(U.conj().T)
    return M

# Perturbations
def random_symmetric_delta(scale=1e-3, complex_phase=False):
    A = np.zeros((3,3), dtype=complex)
    for i in range(3):
        for j in range(i,3):
            mag = np.random.normal(loc=0.0, scale=scale)
            if complex_phase and mag != 0.0:
                phase = np.random.uniform(0,2*np.pi)
                val = mag * np.exp(1j*phase)
            else:
                val = mag
            A[i,j] = val
            A[j,i] = val
```

```
    return A


# Diagonalization
def diagonalize_mass_matrix(M, ordering='normal'):
    w, v = LA.eig(M)
    idx = np.argsort(np.abs(w))
    w_sorted = w[idx]
    v_sorted = v[:, idx]
    for col in range(v_sorted.shape[1]):
        ph = np.angle(v_sorted[0,col])
        v_sorted[:,col] *= np.exp(-1j*ph)
        v_sorted[:,col] /= LA.norm(v_sorted[:,col])
    abs_sorted = np.abs(w_sorted)
    if ordering.lower() == 'normal':
        order_idx = np.argsort(abs_sorted)
    else:
        order_idx = np.argsort(-abs_sorted)
    w_final = w_sorted[order_idx]
    v_final = v_sorted[:, order_idx]
    return w_final, v_final


# Jarlskog
def compute_Jarlskog(U):
    return np.imag(U[0,0]*U[1,1]*np.conj(U[0,1])*np.conj(U[1,0]))


# Extract mixing
def extract_mixing_from_U(U):
    U = np.array(U, dtype=complex)
    s13 = np.abs(U[0,2])
```

```
    c13 = np.sqrt(max(0.0, 1.0 - s13**2))
    s12 = np.abs(U[0,1]) / (c13 + 1e-16)
    s23 = np.abs(U[1,2]) / (c13 + 1e-16)
    s12 = clamp(s12, 0.0, 1.0)
    s23 = clamp(s23, 0.0, 1.0)
    c12 = np.sqrt(max(0.0, 1.0 - s12**2))
    c23 = np.sqrt(max(0.0, 1.0 - s23**2))
    J = compute_Jarlskog(U)
    denom = s12 * s23 * s13 * c12 * c23 * (c13**2)
    if abs(denom) < 1e-20:
        sin_delta = 0.0
    else:
        sin_delta = clamp(J / denom, -1.0, 1.0)
    delta_rad = np.arcsin(sin_delta)
    theta12_deg = rad2deg(np.arcsin(s12))
    theta13_deg = rad2deg(np.arcsin(s13))
    theta23_deg = rad2deg(np.arcsin(s23))
    delta_deg = rad2deg(delta_rad)
    return {
        'theta12_deg': theta12_deg,
        'theta13_deg': theta13_deg,
        'theta23_deg': theta23_deg,
        'delta_deg_est': delta_deg,
        'J': J
    }

# Monte Carlo scan
def monte_carlo_scan(theta12,theta13,theta23,delta,masses,
                     epsilon_list, n_samples_per_epsilon=200,
```

```
                    random_scale=1e-3, complex_phase=False,
                    mass_matrix_type='Majorana', ordering='normal'):
    results = {'epsilon': [], 'mean_delta_change_deg': [], 'std_delta_change_deg': []
               'mean_J_change': [], 'std_J_change': [], 'all_delta_changes': [], 'all
    for eps in tqdm(epsilon_list):
        delta_changes = []
        J_changes = []
        for _ in range(n_samples_per_epsilon):
            dM = random_symmetric_delta(scale=random_scale, complex_phase=complex_pha
            U0 = build_PMNS(theta12,theta13,theta23,delta)
            M0 = build_mass_matrix_from_masses(U0, masses, mass_matrix_type=mass_matr
            M1 = M0 + eps * dM
            w1, v1 = diagonalize_mass_matrix(M1, ordering=ordering)
            params1 = extract_mixing_from_U(v1)
            delta_changes.append(params1['delta_deg_est'] - delta)
            J_changes.append(params1['J'] - compute_Jarlskog(U0))
        results['epsilon'].append(eps)
        results['mean_delta_change_deg'].append(np.mean(delta_changes))
        results['std_delta_change_deg'].append(np.std(delta_changes))
        results['mean_J_change'].append(np.mean(J_changes))
        results['std_J_change'].append(np.std(J_changes))
        results['all_delta_changes'].append(delta_changes)
        results['all_J_changes'].append(J_changes)
    return results

# Main function
def main(args):
    nufit = {'theta12_deg':33.44,'theta13_deg':8.57,'theta23_deg':49.2,'delta_cp_deg'
             'dm21_sq':7.42e-5,'dm31_sq':2.517e-3,'m1_ev':0.001}
```

```
    theta12 = nufit['theta12_deg']
    theta13 = nufit['theta13_deg']
    theta23 = nufit['theta23_deg']
    delta_cp = nufit['delta_cp_deg']
    m1 = nufit['m1_ev']
    m2 = np.sqrt(m1**2 + nufit['dm21_sq'])
    m3 = np.sqrt(m1**2 + nufit['dm3l_sq'])
    masses = [m1,m2,m3]
    results = monte_carlo_scan(theta12,theta13,theta23,delta_cp,masses,
                               args.epsilon_list, n_samples_per_epsilon=args.n_sample
                               random_scale=args.random_scale, complex_phase=args.com
                               mass_matrix_type=args.mass_matrix_type, ordering=args.
    print("Scan complete. Summary:")
    for i, eps in enumerate(results['epsilon']):
        print(f"epsilon={eps}, mean delta change={results['mean_delta_change_deg'][i]

main(args)
```

Notes and Remarks

- Ensure `numpy`, `scipy`, `matplotlib`, `pandas`, and `tqdm` are installed.
- The code runs Monte Carlo perturbations for each ϵ in `epsilon_list` and outputs histograms, summary CSVs, and plots.
- Physicality and unitarity checks are implemented to flag invalid samples.
- Adjust `Args` parameters for larger sample sizes or different perturbation scales.

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